STAGE 2 — COMPREHENSIVE

ember_gbr

Order 2 | metric / inter-timestep / aggregates

```
T (records)         : 26
D (carriers)        : 9
Carriers            : Bioenergy, Coal, Gas, Hydro, Nuclear, Other Fossil, Other Renewables, Solar, Win
Engine              : cnt 2.0.4
Schema              : 2.1.0
Source SHA-256      : b1df2a0ae6f21f77c0d6fb5b7a0c26af...
Content SHA-256     : 56ced72dc4e2fb295b563614fe51b17e...
```

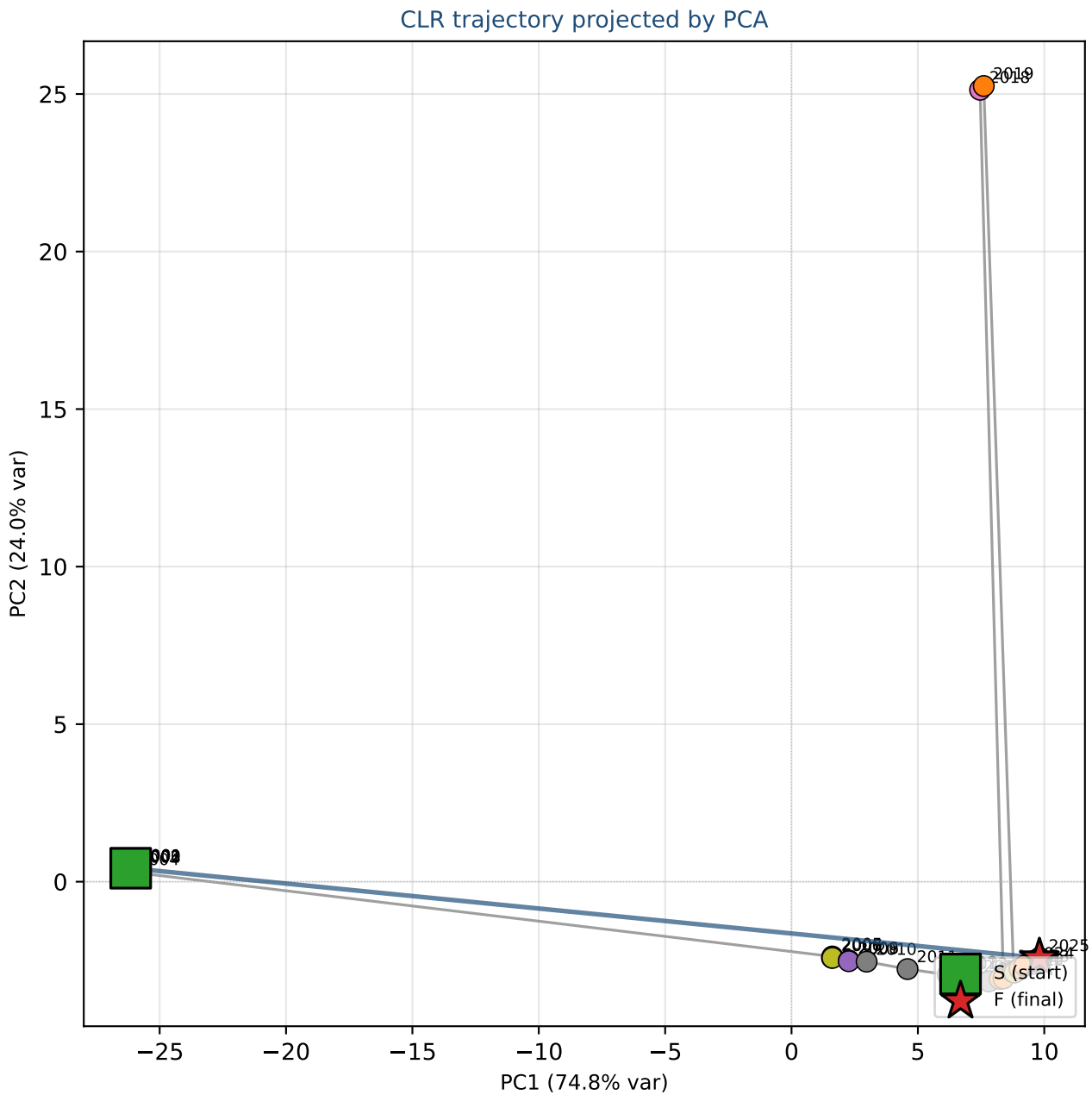
Doctrine: integer orders only. Order 1 = single composition.
Order 2 = anything aggregating across multiple timesteps.

This Stage 2 atlas covers:

- Trajectory aggregates (Hs, Aitchison-norm, ILR-axes)
- Inter-timestep metrics (ω, κ, helmsman, ring, locks)
- Pairwise structure (variation, correlation, divergence)
- System Course Plot (CNT Math Handbook Ch 15)
- Compositional Bearing Scope CBS (Ch 16)
- Numeric summary

STAGE 2 | Plate 2 | System Course Plot — PCA 2D path with S→F net vector

ember_gbr | Engine cnt 2.0.4 | Schema 2.1.0 | Order 2

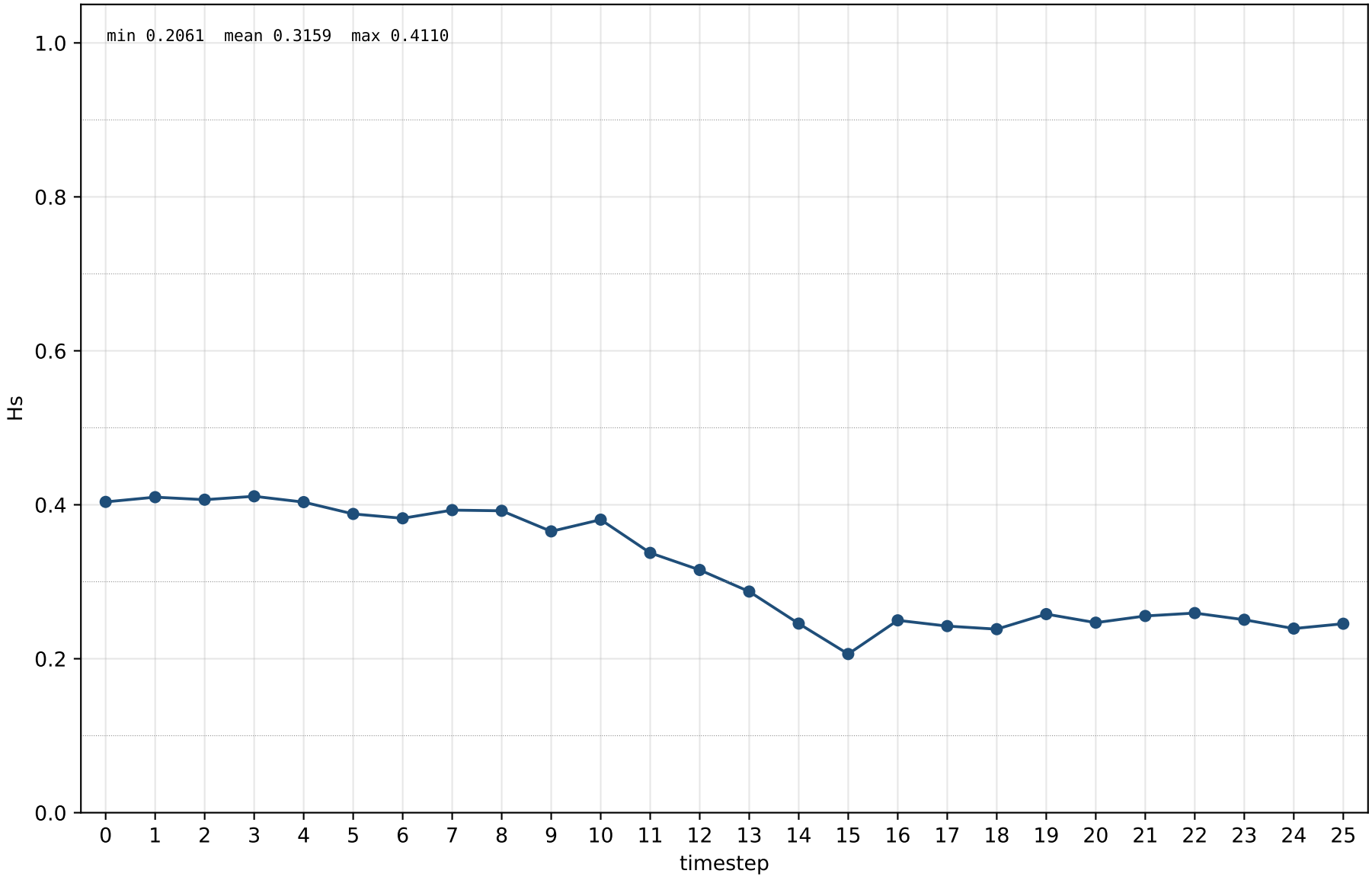


Navigation metrics (V_net = h_F − h_S)

```
h_start label    : 2000

h_final label    : 2025


net_distance     : 36.2591 HLR

path_length      : 100.3591 HLR

course_directness: 0.3613

  (1.0 = straight line, 0 = pure looping)


dynamic range S  : 39.5365 HLR

dynamic range F  : 39.0486 HLR


Reading guide:

  tight cluster  → stable composition

  long excursion → structural shift

  sharp turn     → regime transition

  loop           → cyclic behaviour

  spiral         → systematic drift
```

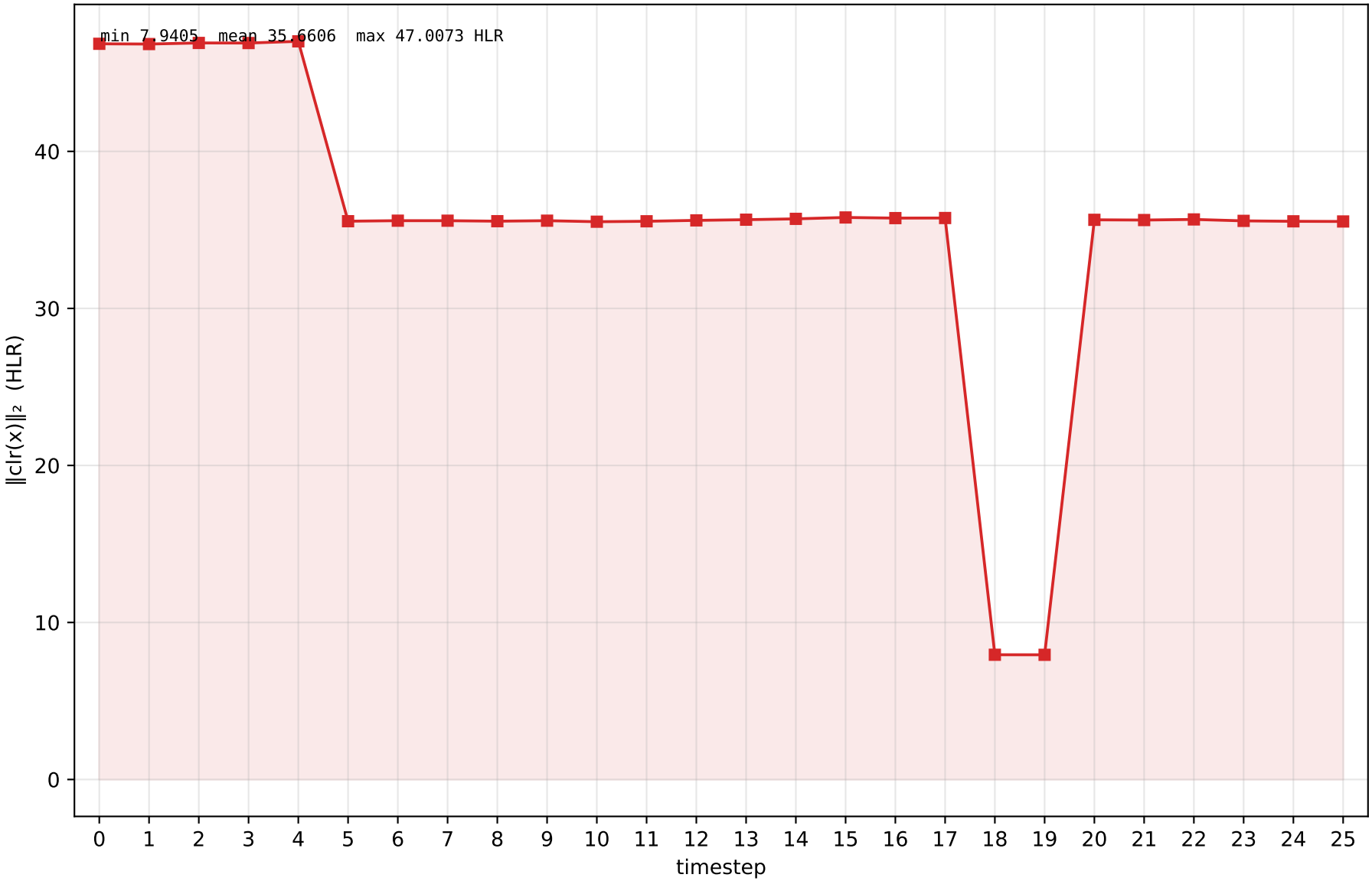
STAGE 2 | Plate 3 | Hs (Higgins scale) trajectory — Order 2 aggregate

ember_gbr | Engine cnt 2.0.4 | Schema 2.1.0 | Order 2



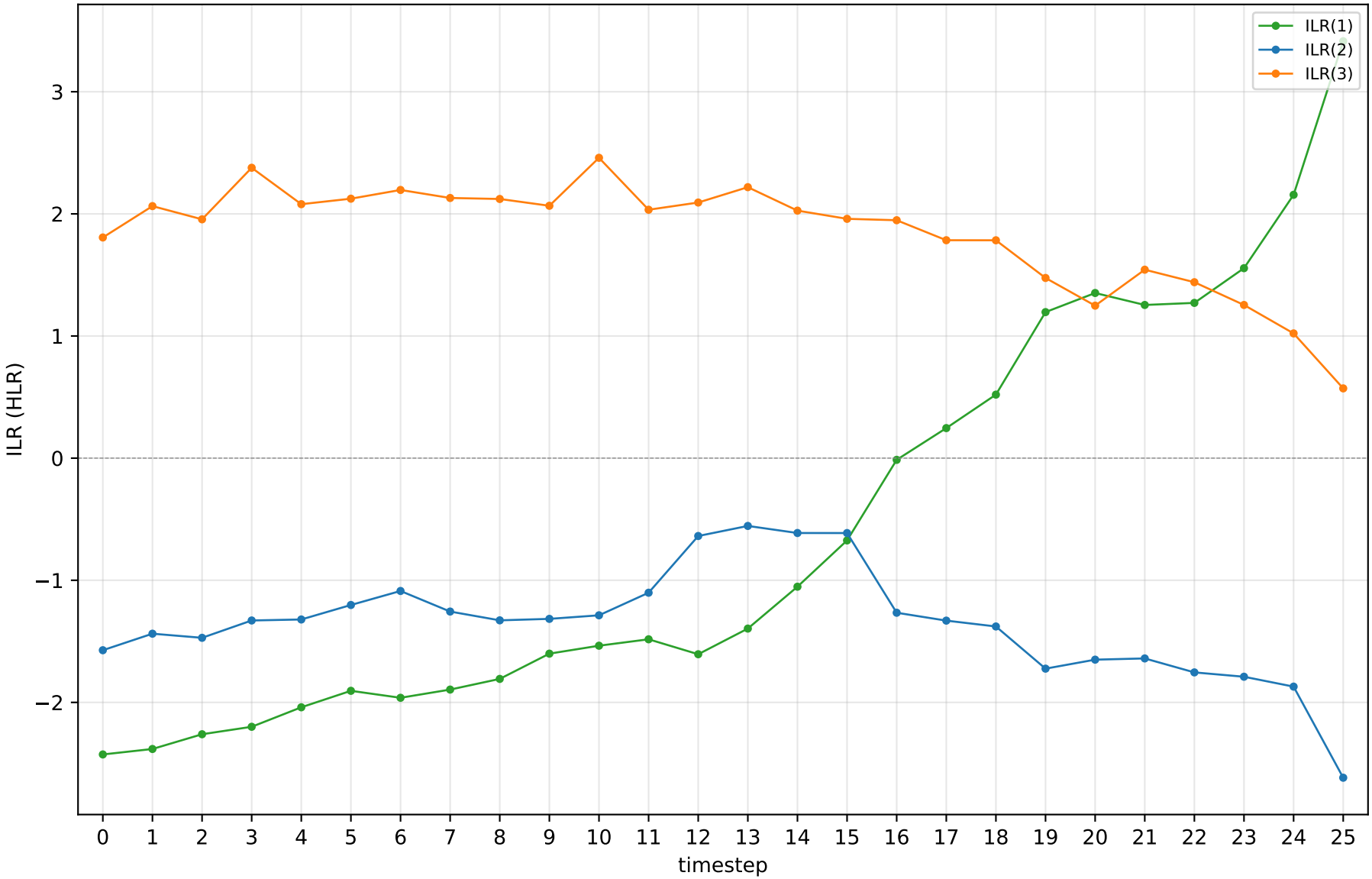
STAGE 2 | Plate 4 | Aitchison norm trajectory ‖clr(x)‖₂ — Order 2 aggregate

ember_gbr | Engine cnt 2.0.4 | Schema 2.1.0 | Order 2



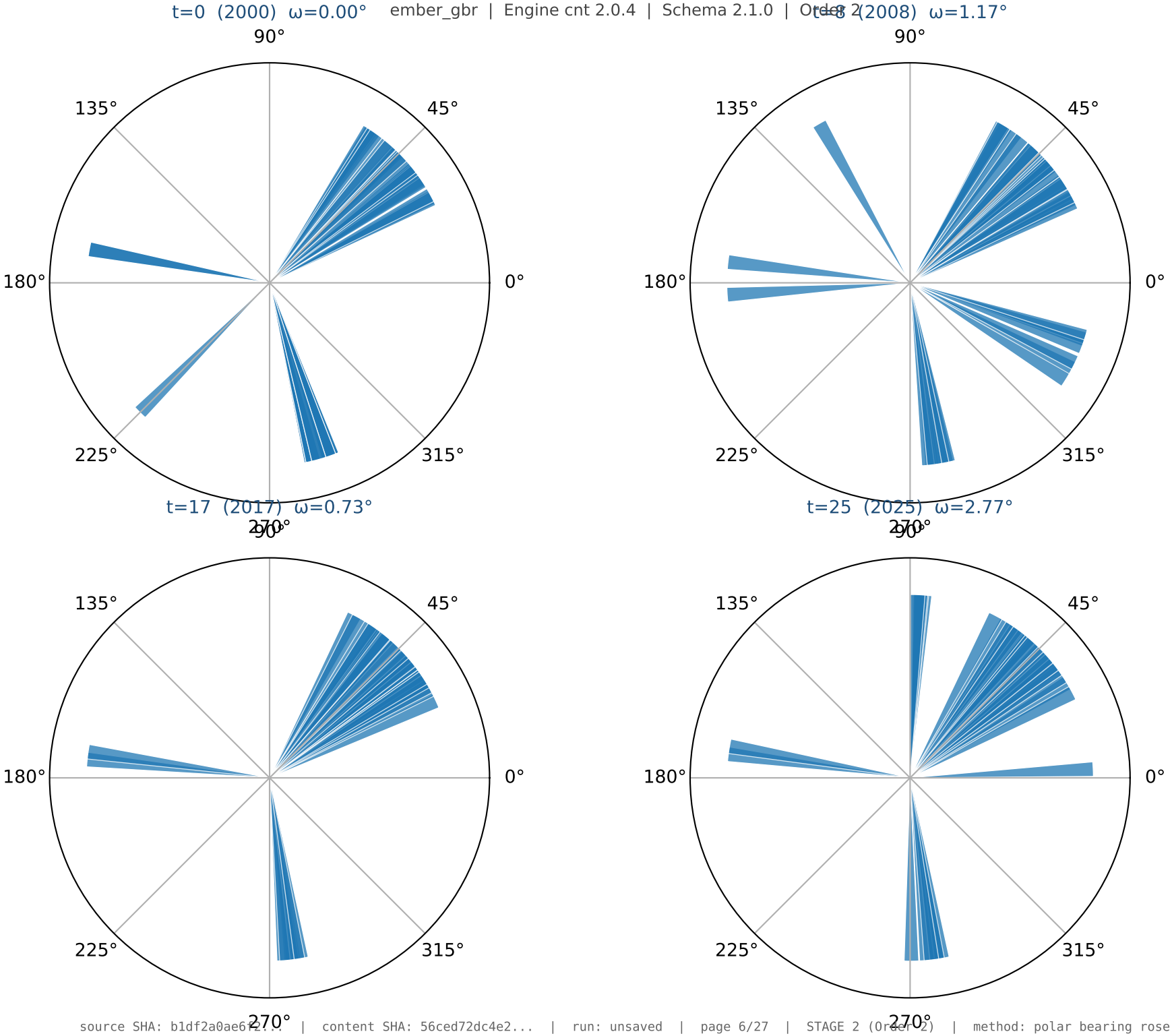
STAGE 2 | Plate 5 | ILR axes 1–3 trajectories (Helmert basis) — Order 2 aggregate

ember_gbr | Engine cnt 2.0.4 | Schema 2.1.0 | Order 2



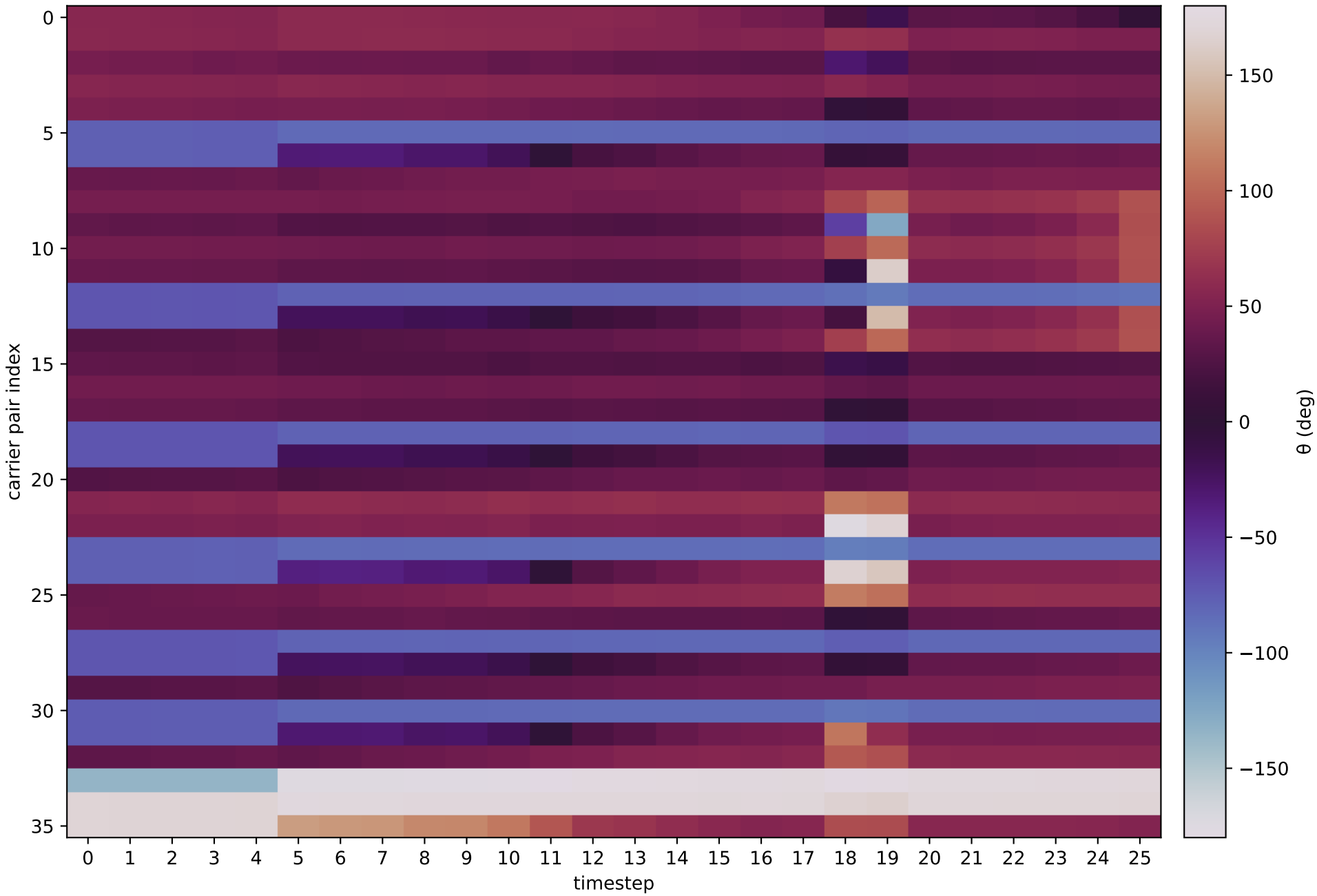
STAGE 2 | Plate 6 | Polar bearing rose (4 sample timesteps)

ember_gbr | Engine cnt 2.0.4 | Schema 2.1.0 | Order 2



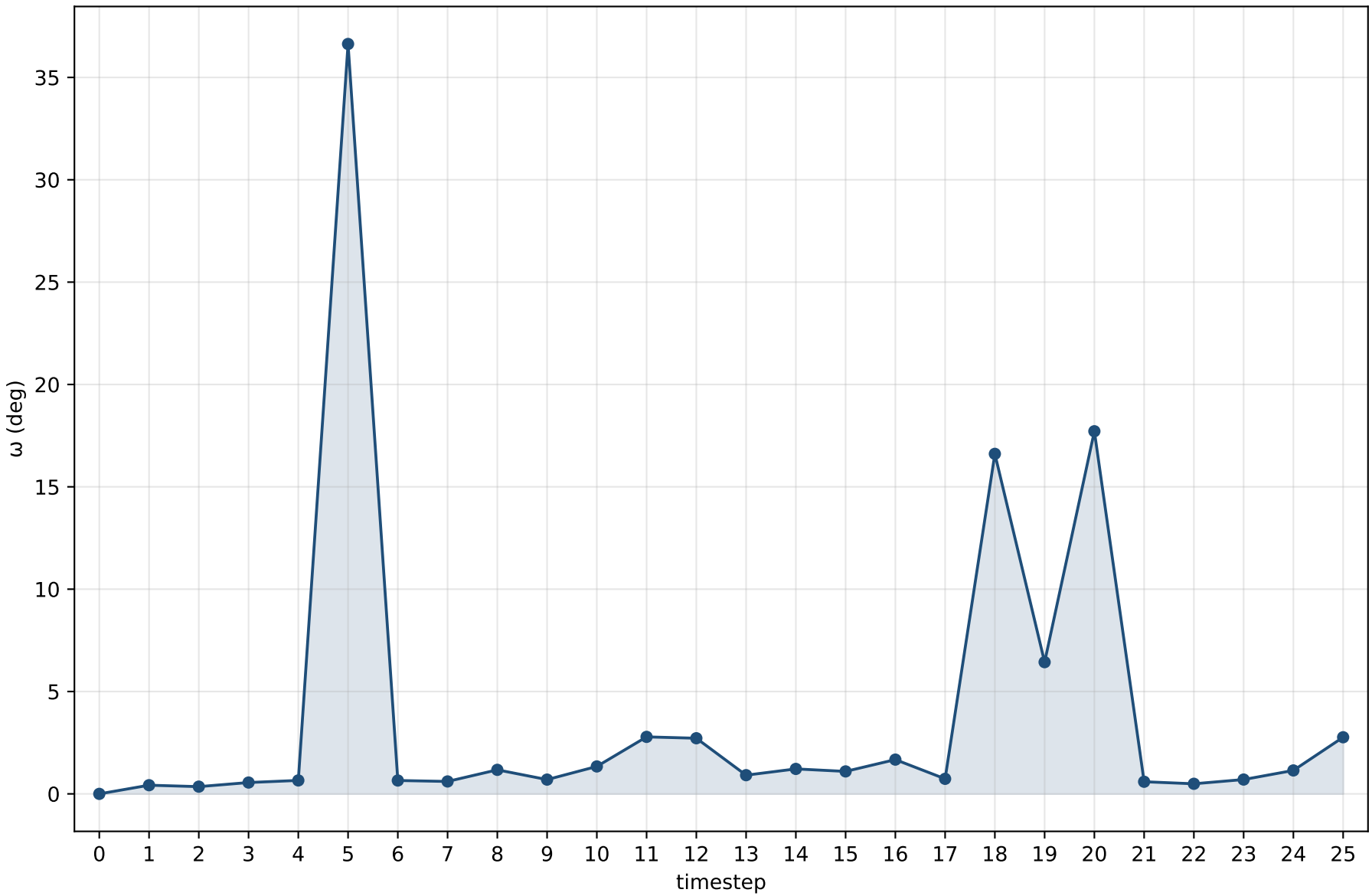
STAGE 2 | Plate 7 | Bearing heatmap T × C(D,2) pairs

ember_gbr | Engine cnt 2.0.4 | Schema 2.1.0 | Order 2



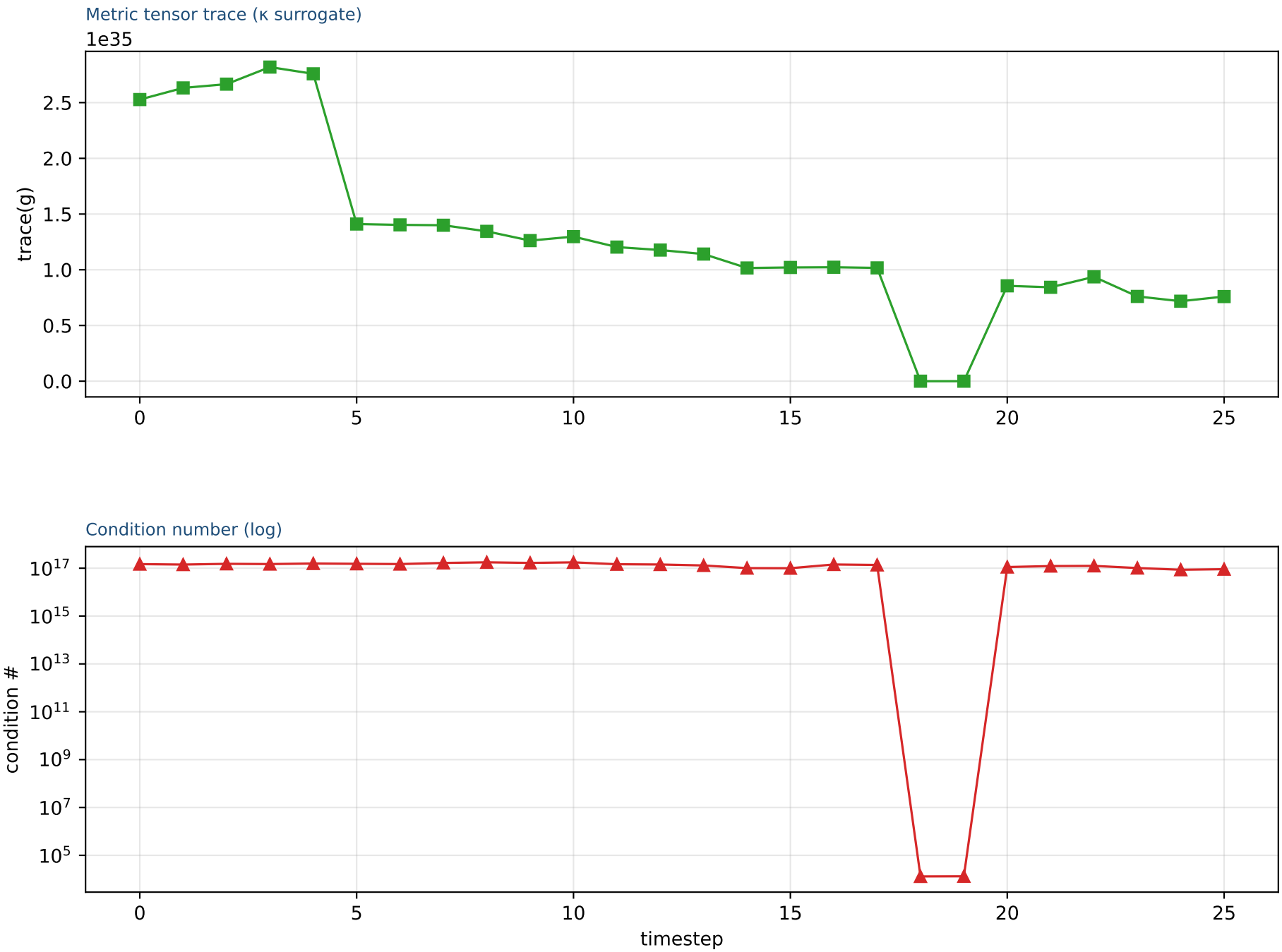
STAGE 2 | Plate 8 | ω angular velocity trajectory

ember_gbr | Engine cnt 2.0.4 | Schema 2.1.0 | Order 2



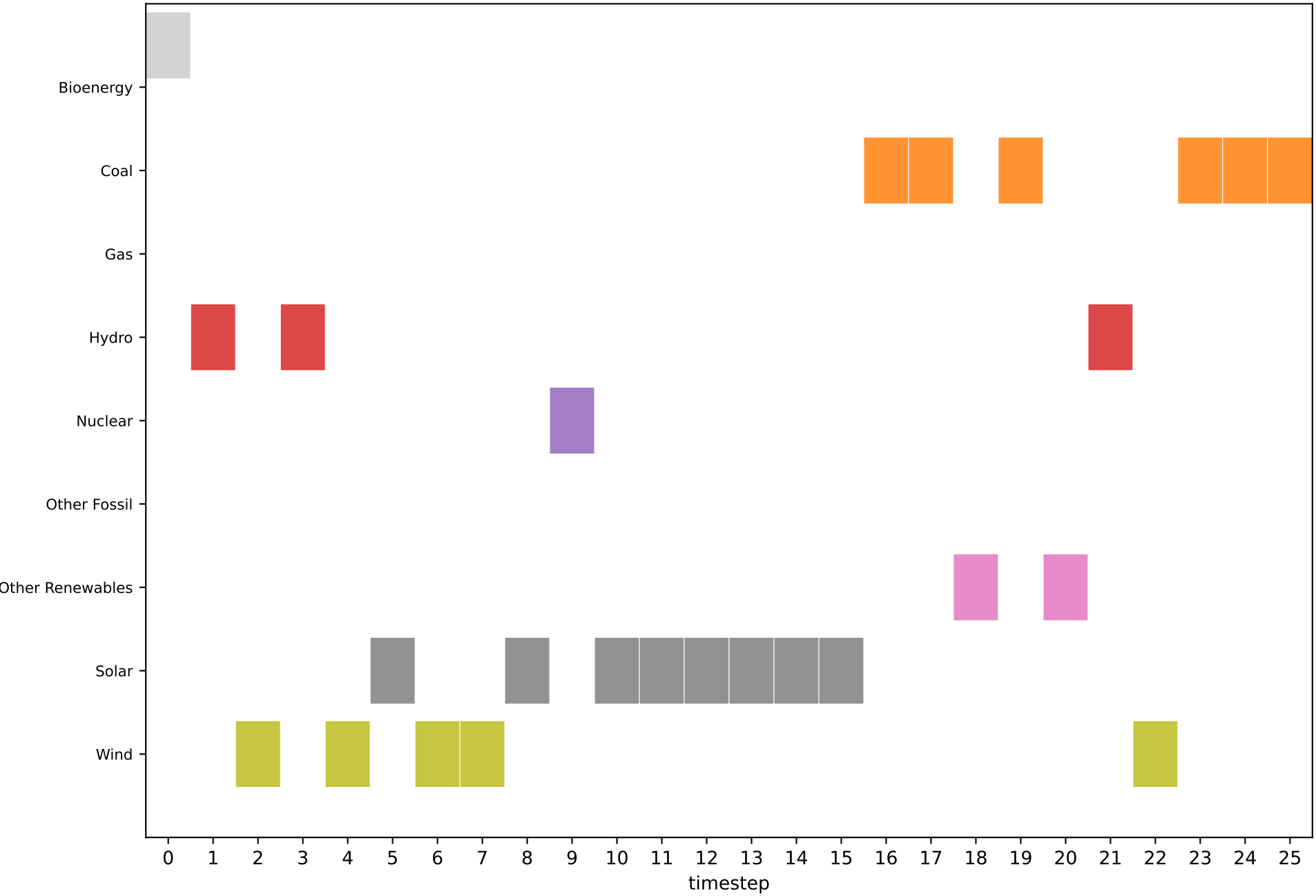
STAGE 2 | Plate 9 | Metric tensor trace + condition number

ember_gbr | Engine cnt 2.0.4 | Schema 2.1.0 | Order 2



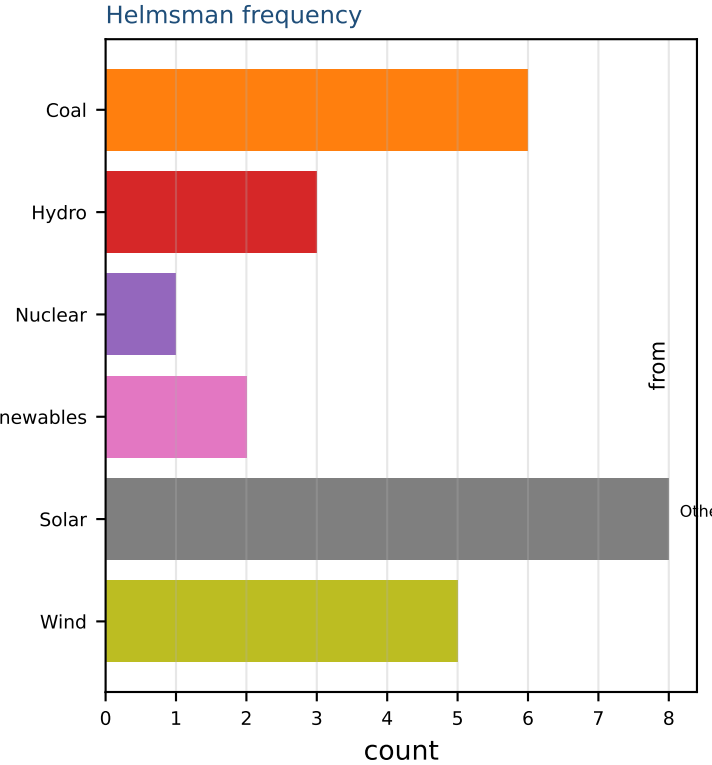
STAGE 2 | Plate 10 | Helmsman σ lineage timeline

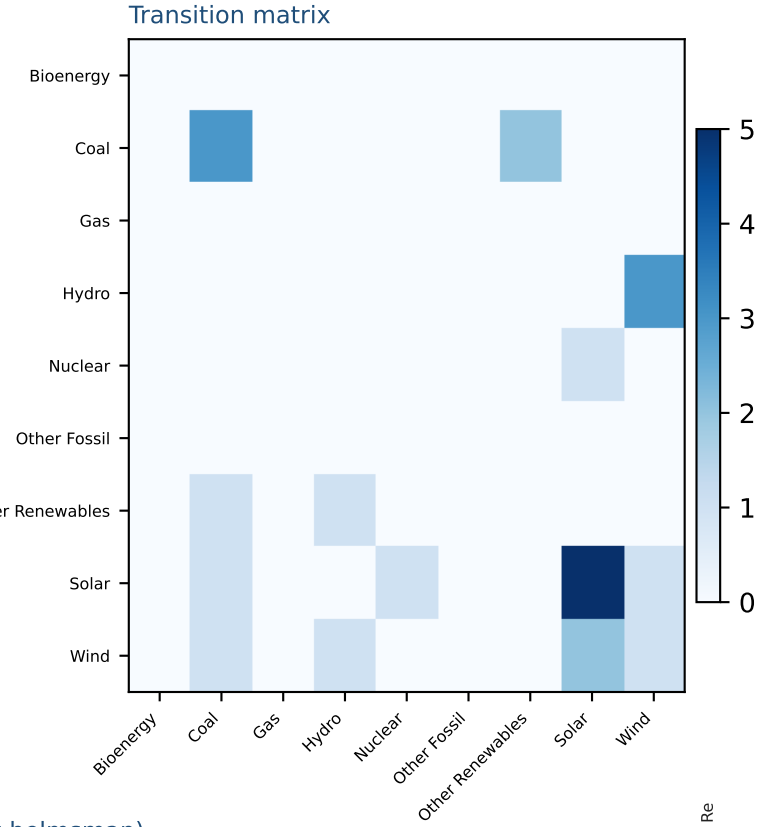
ember_gbr | Engine cnt 2.0.4 | Schema 2.1.0 | Order 2

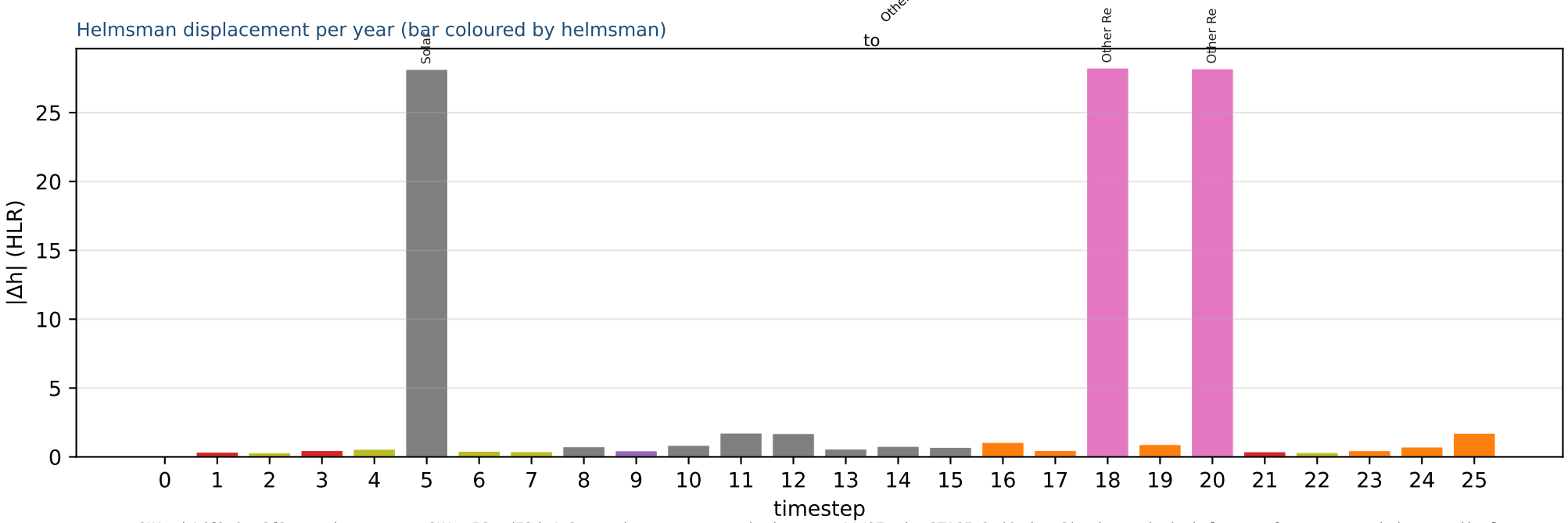


STAGE 2 | Plate 11 | Helmsman σ — frequency · transitions · per-year displacement

ember_gbr | Engine cnt 2.0.4 | Schema 2.1.0 | Order 2

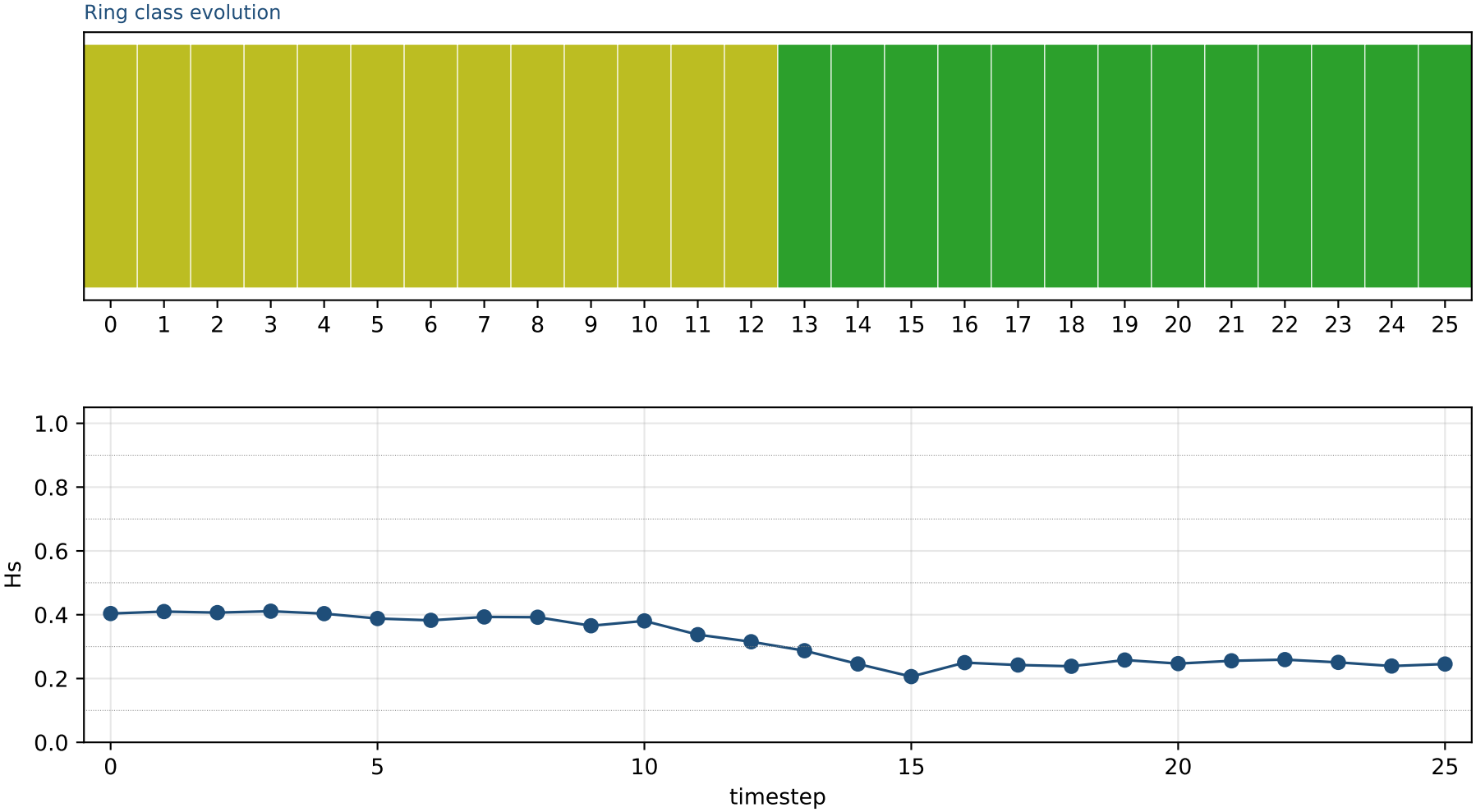






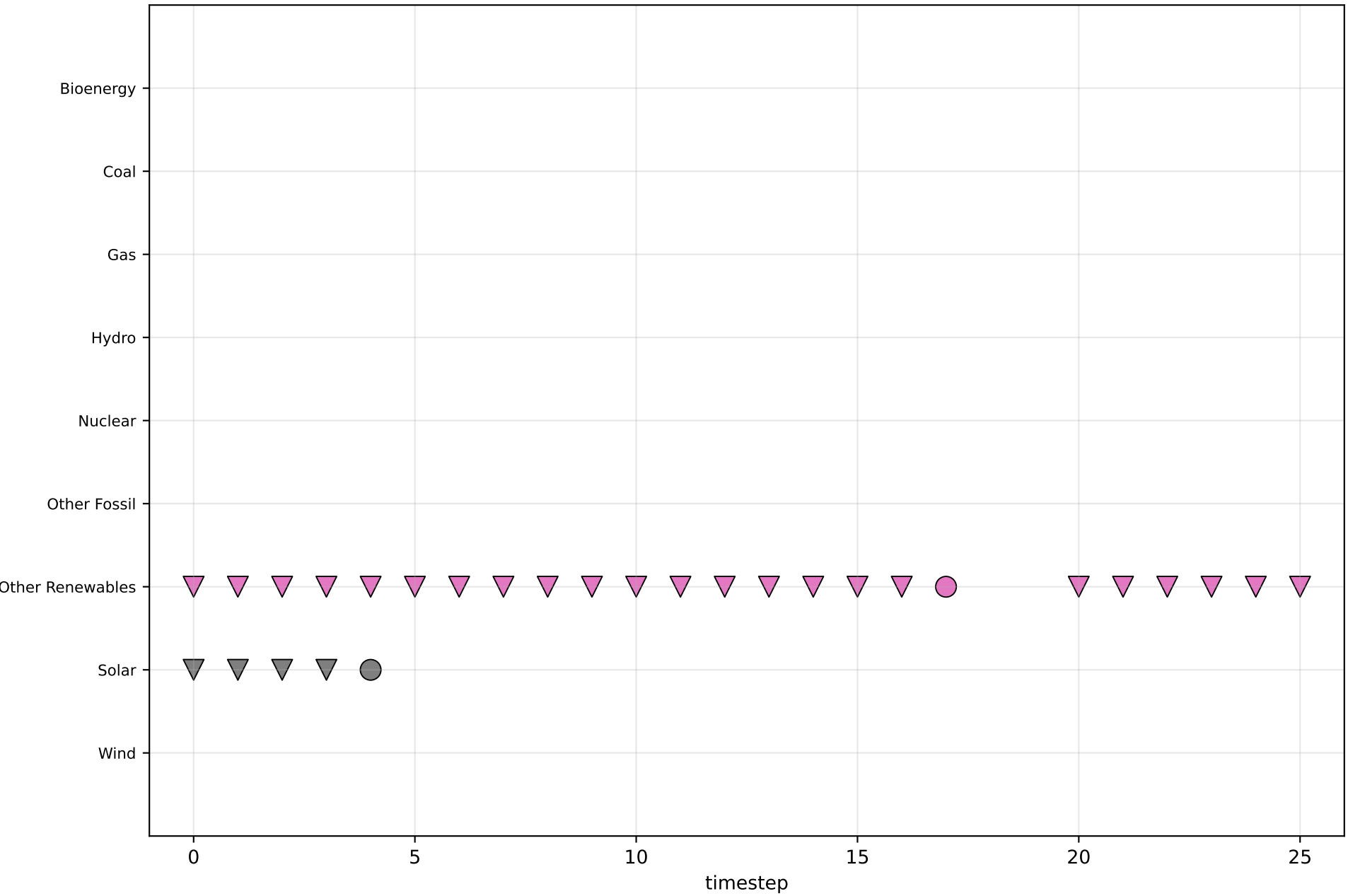
STAGE 2 | Plate 12 | Ring class evolution + Hs

ember_gbr | Engine cnt 2.0.4 | Schema 2.1.0 | Order 2



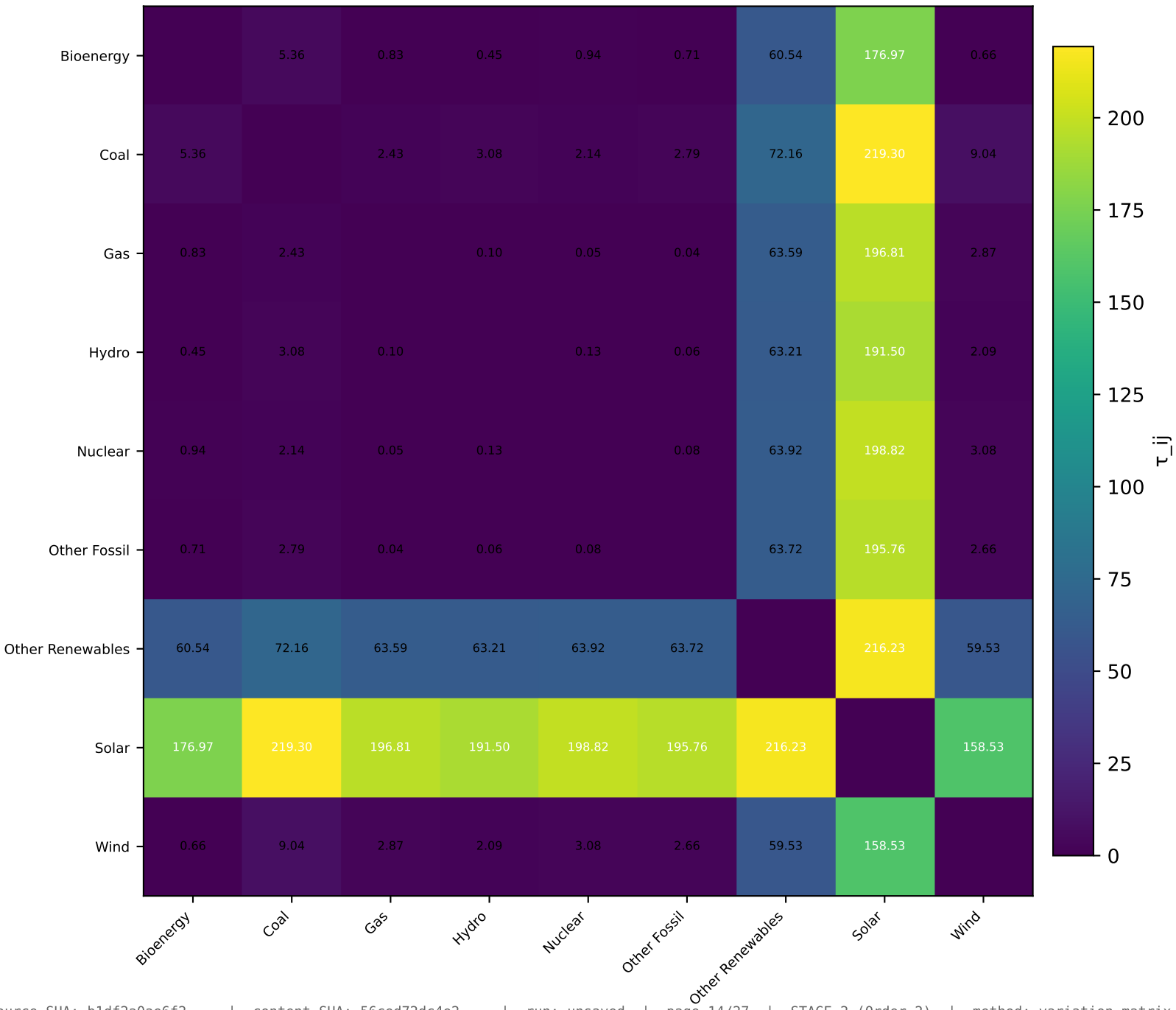
STAGE 2 | Plate 13 | Lock event timeline (carriers × timesteps)

ember_gbr | Engine cnt 2.0.4 | Schema 2.1.0 | Order 2



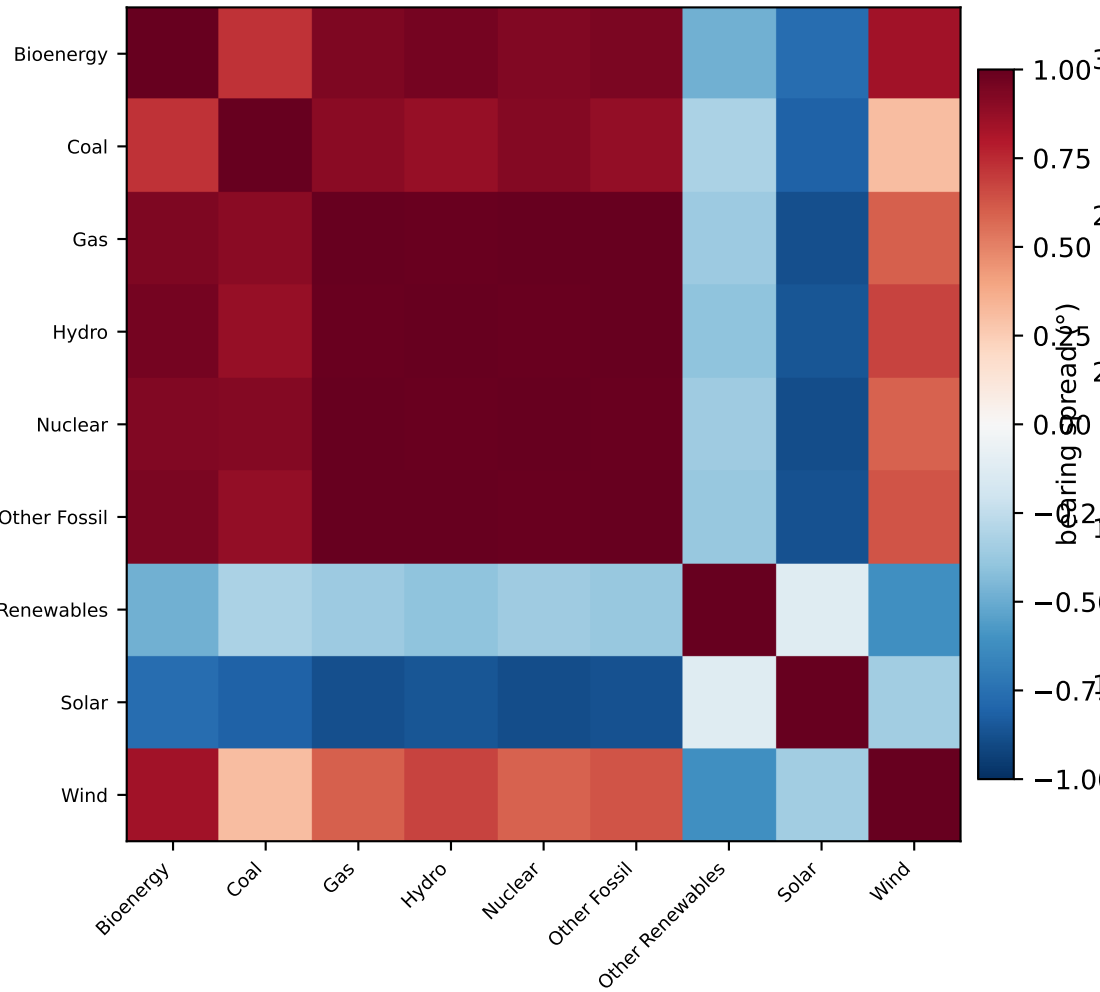
STAGE 2 | Plate 14 | Variation matrix τ_ij heatmap

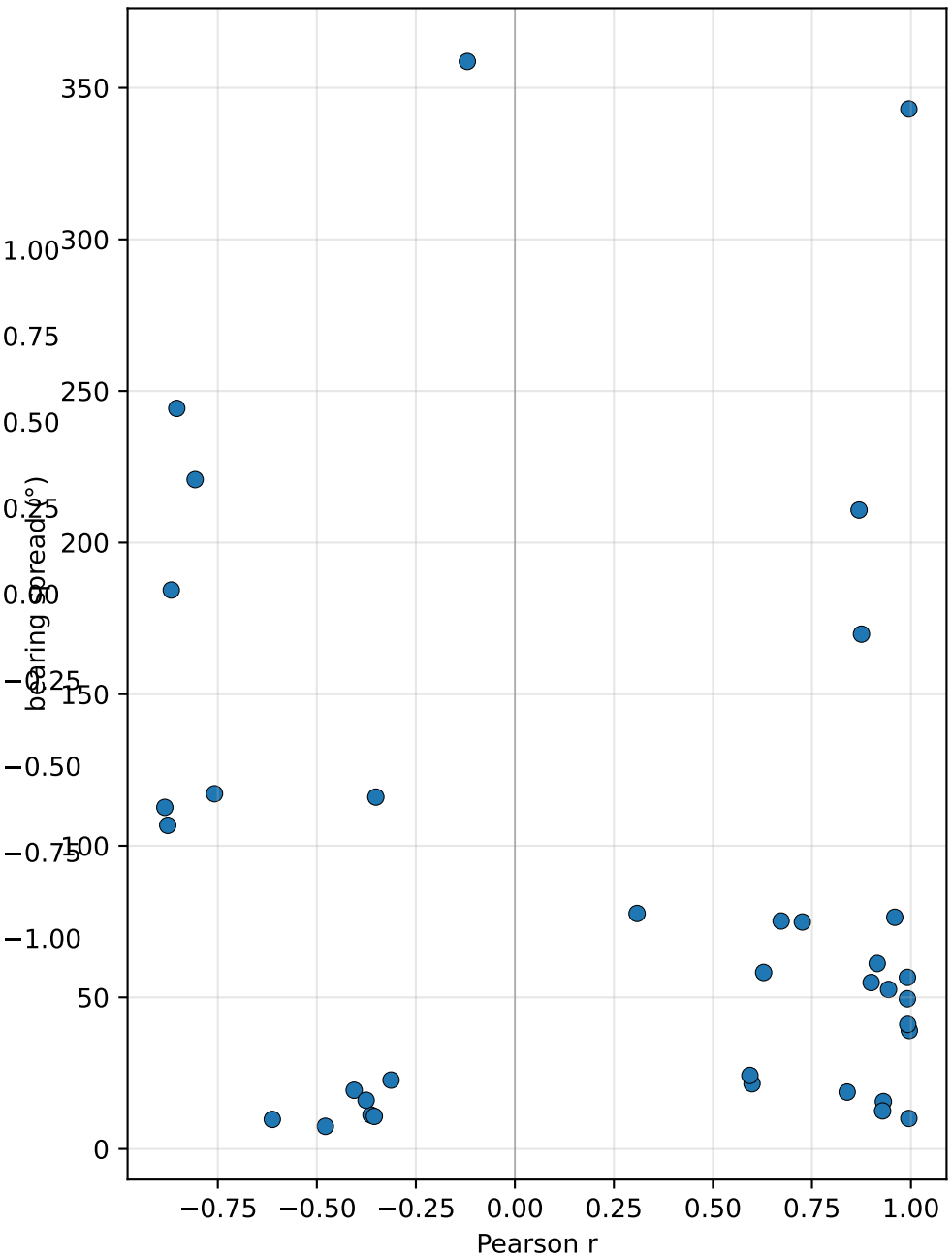
ember_gbr | Engine cnt 2.0.4 | Schema 2.1.0 | Order 2



STAGE 2 | Plate 15 | Pair Pearson r heatmap + co-movement scatter

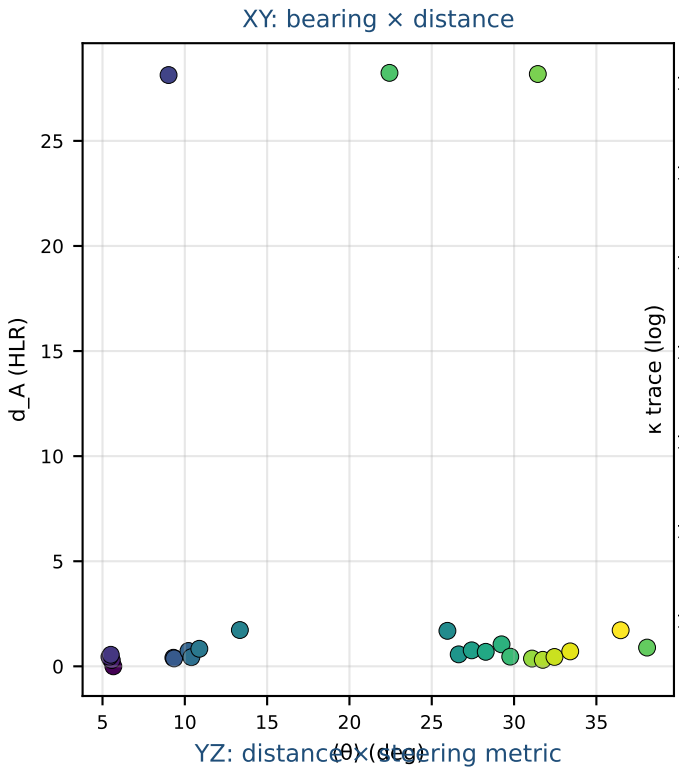
ember_gbr | Engine cnt 2.0.4 | Schema 2.1.0 | Order 2

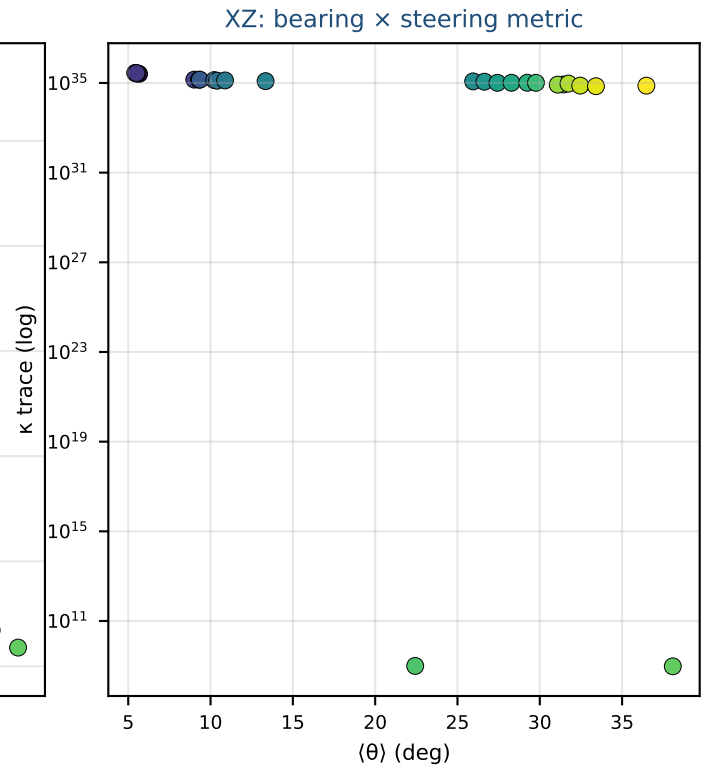


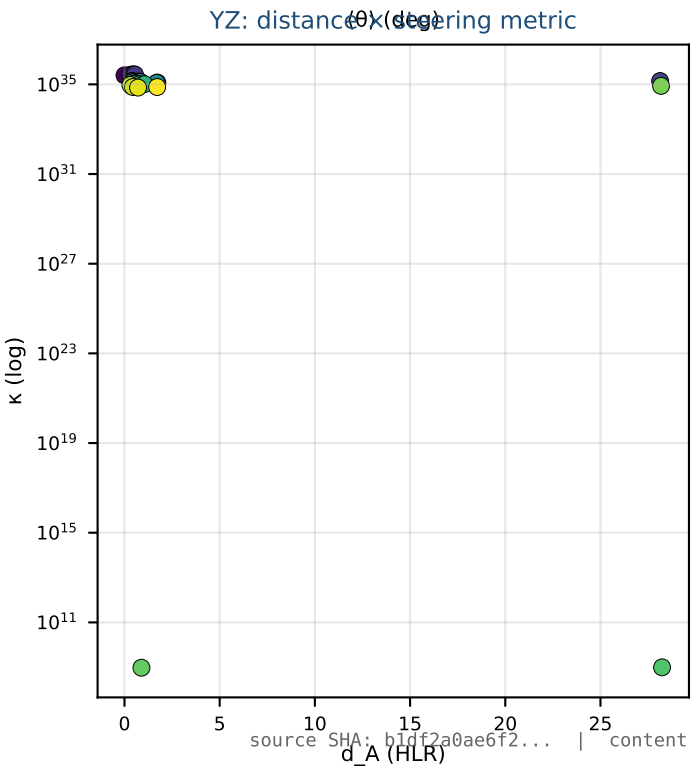


STAGE 2 | Plate 16 | Compositional Bearing Scope (CBS) — 3 orthogonal faces (Math Handbook Ch 16)

ember_gbr | Engine cnt 2.0.4 | Schema 2.1.0 | Order 2





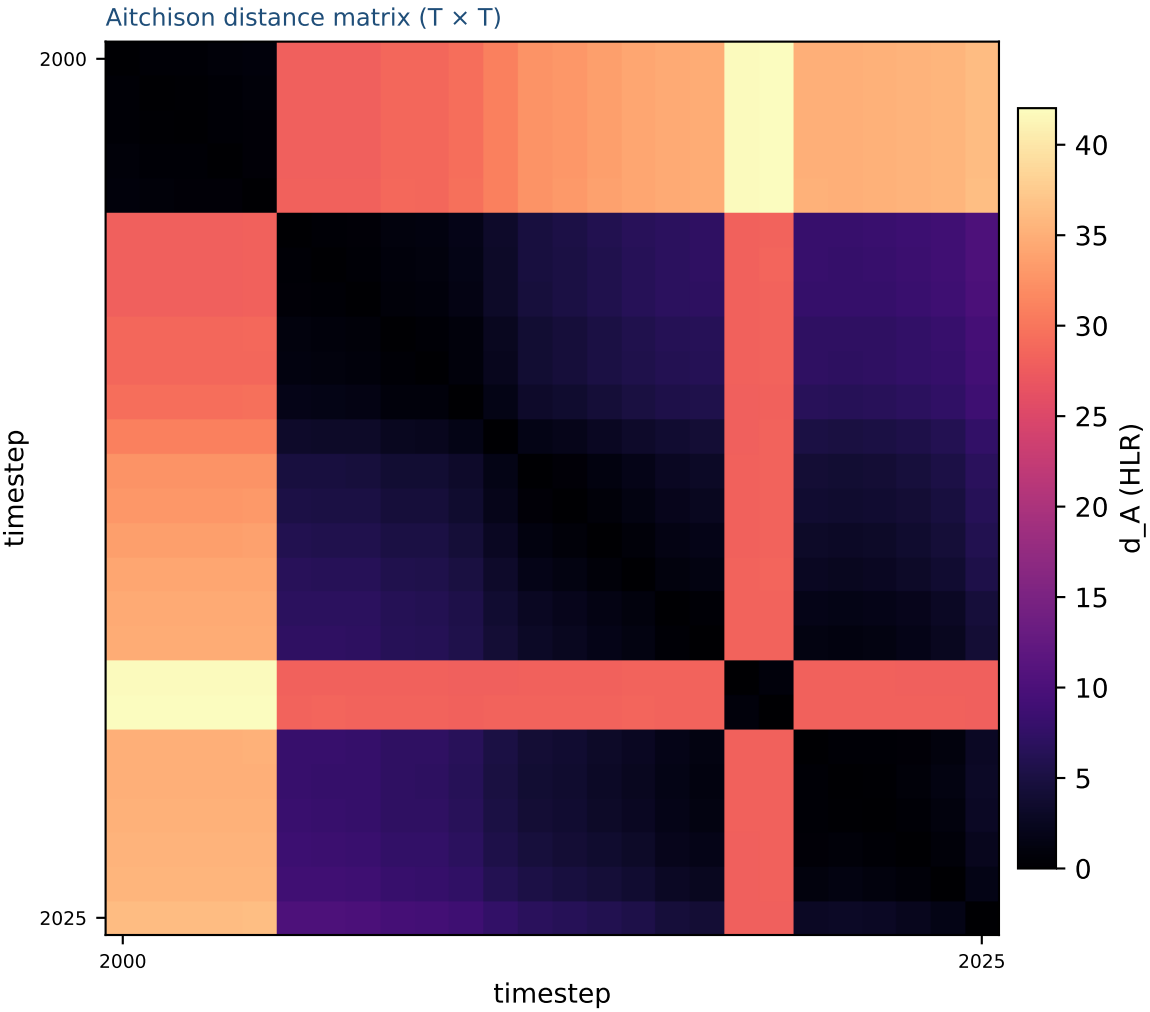


CBS info panel (Math Handbook Ch 16)

```
⟨θ⟩ range  : [5.4, 38.1]°
d_A range  : [0.0000, 28.2360] HLR
κ range    : [9.52e+08, 2.82e+35]
Hs range   : [0.2061, 0.4110]
ω max      : 36.63°

Each face: one observation per timestep.
Colour = timestep (early=dark, late=bright).
Three faces locate the system in the
(bearing, distance, steering) measurement cube.
```

STAGE 2 | Plate 17 | Pairwise divergence — distance matrix + top-15 ranking

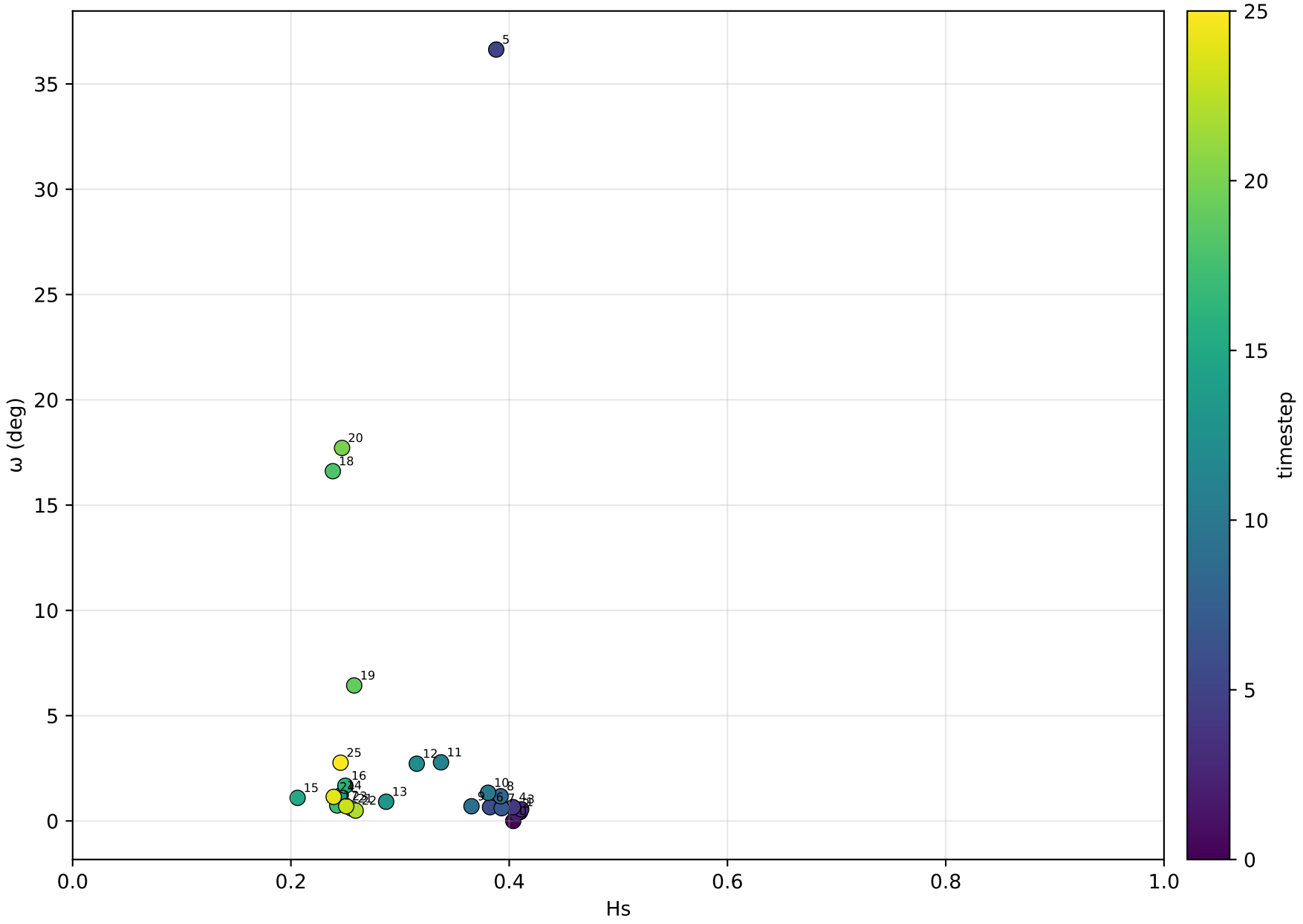
ember_gbr | Engine cnt 2.0.4 | Schema 2.1.0 | Order 2



rank	A	B	d_A
1	2004	2019	42.0223
2	2002	2019	41.9369
3	2003	2019	41.9196
4	2000	2019	41.9006
5	2001	2019	41.8750
6	2004	2018	41.8489
7	2002	2018	41.7611
8	2003	2018	41.7407
9	2000	2018	41.7240
10	2001	2018	41.6954
11	2004	2025	36.2846
12	2002	2025	36.2613
13	2000	2025	36.2591
14	2003	2025	36.2581
15	2001	2025	36.2497

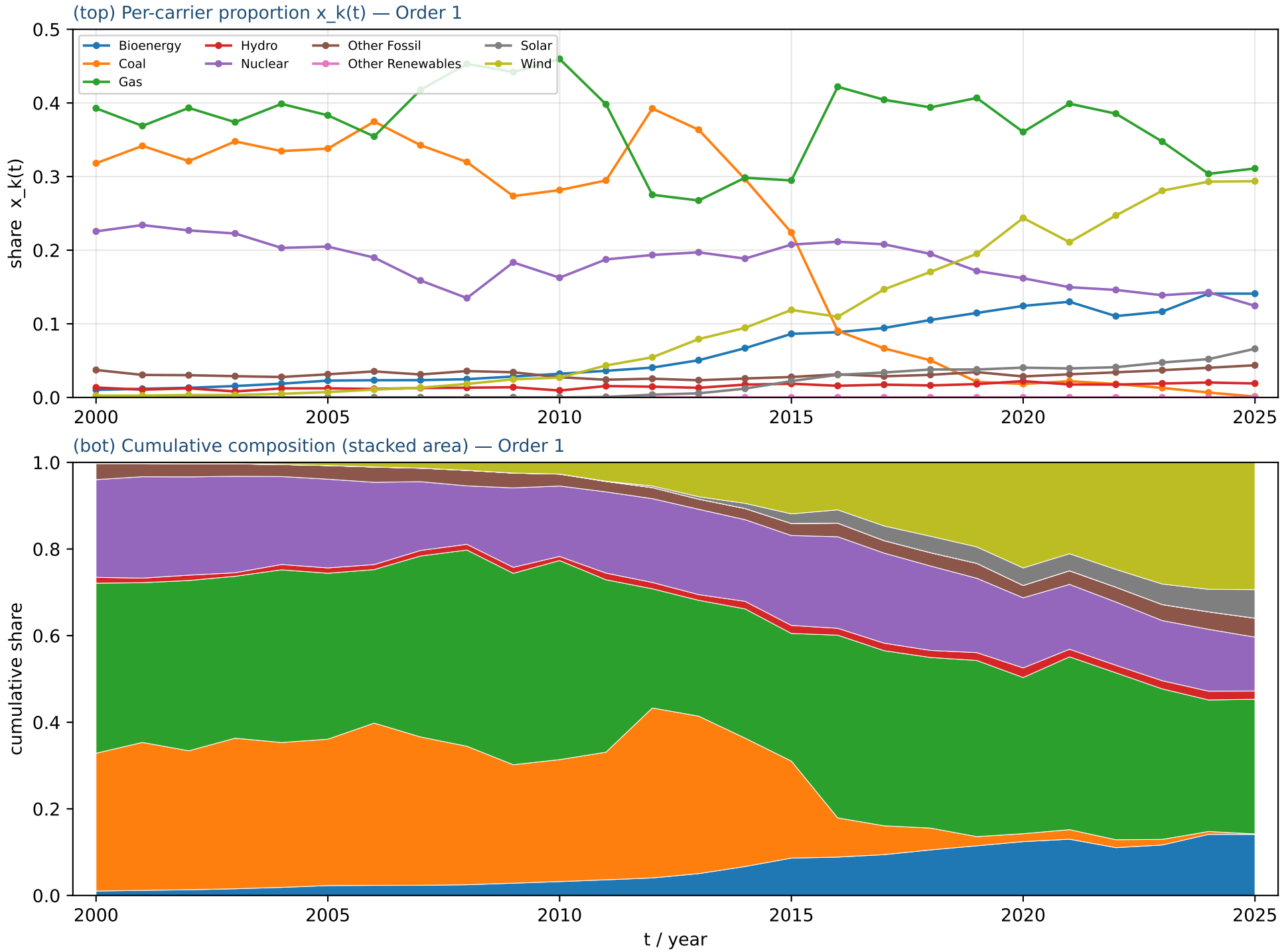
STAGE 2 | Plate 18 | ω vs Hs scatter (XY test)

ember_gbr | Engine cnt 2.0.4 | Schema 2.1.0 | Order 2



STAGE 2 | Plate 19 | Evolution of Proportions (raw + cumulative)

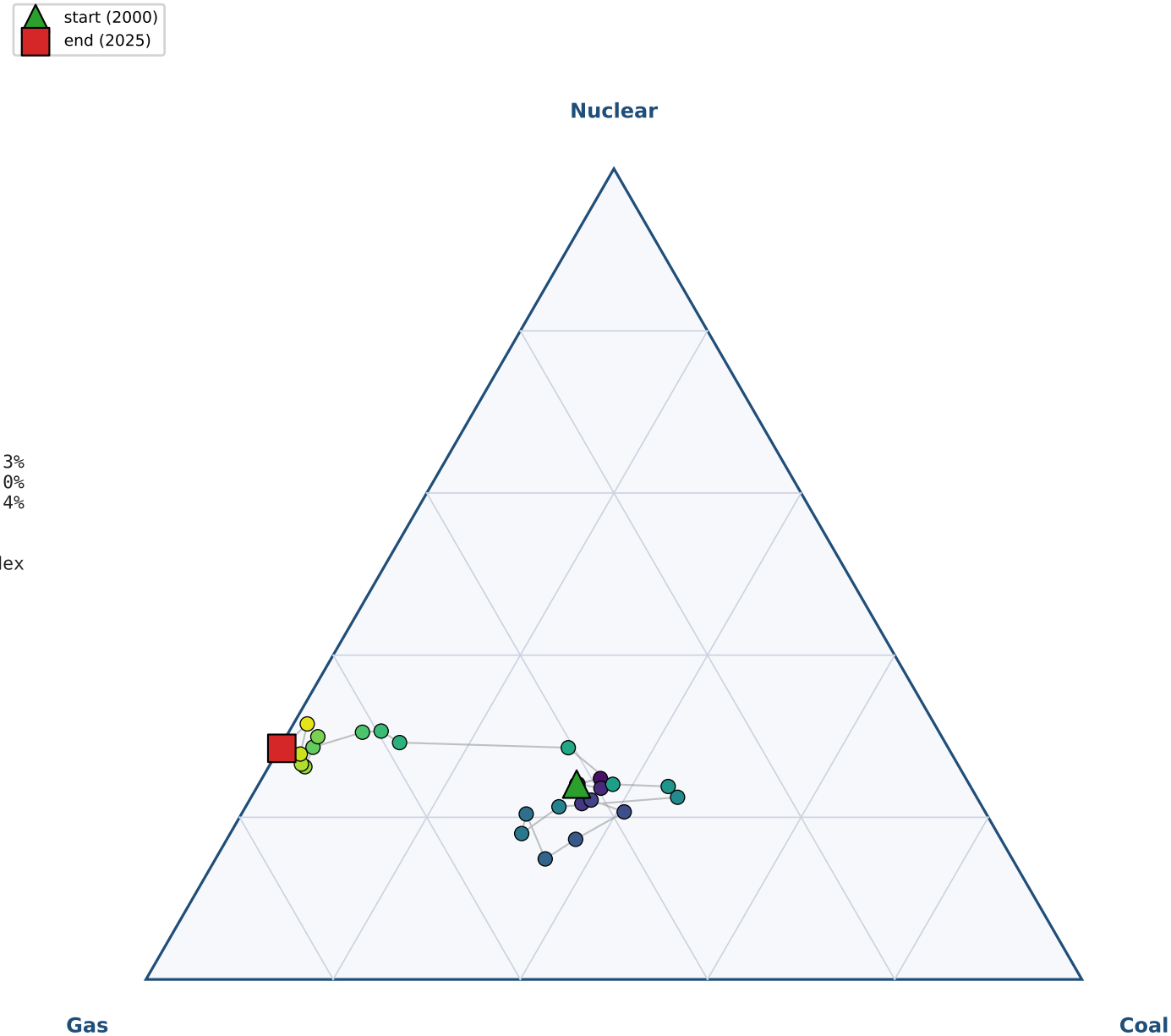
ember_gbr | Engine cnt 2.0.4 | Schema 2.1.0 | Order 2



STAGE 2 | Plate 20 | Ternary Diagram (top-3 carriers)

ember_gbr | Engine cnt 2.0.4 | Schema 2.1.0 | Order 2

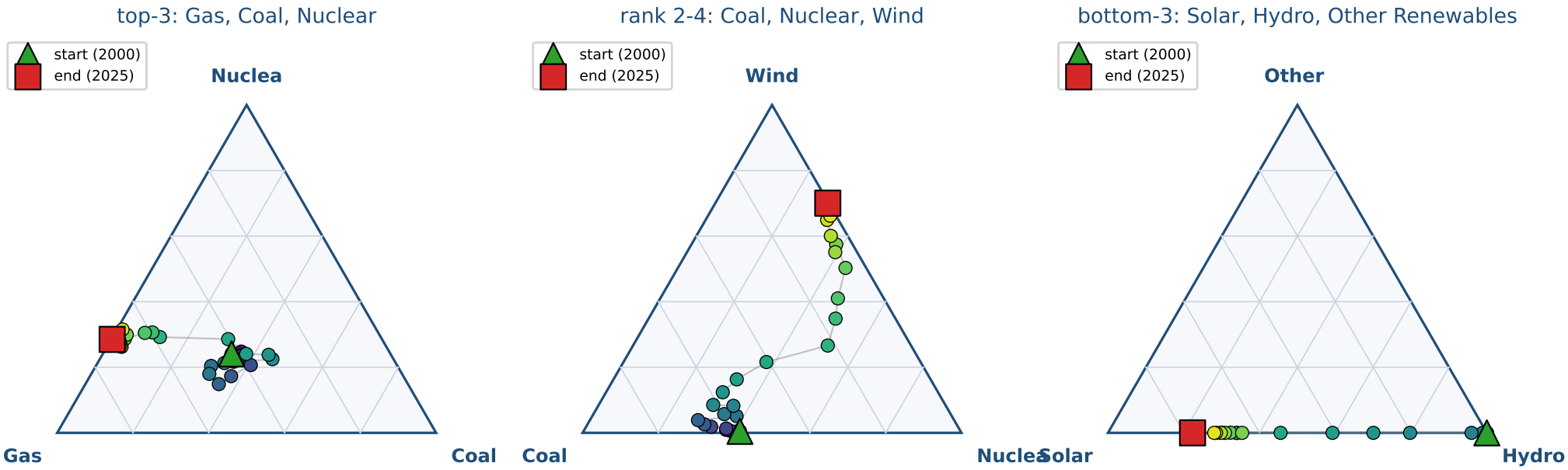
3-part subcomposition closed over Gas, Coal, Nuclear
(remaining 6 carriers absorbed into closure)




```
Top-3 by mean share:
  Gas             37.3%
  Coal            21.0%
  Nuclear         18.4%

T = 26 timesteps
colour map = time index
▲ start  ■ end
```

STAGE 2 | Plate 21 | Three-Part Subcompositions (3 ternaries)

ember_gbr | Engine cnt 2.0.4 | Schema 2.1.0 | Order 2

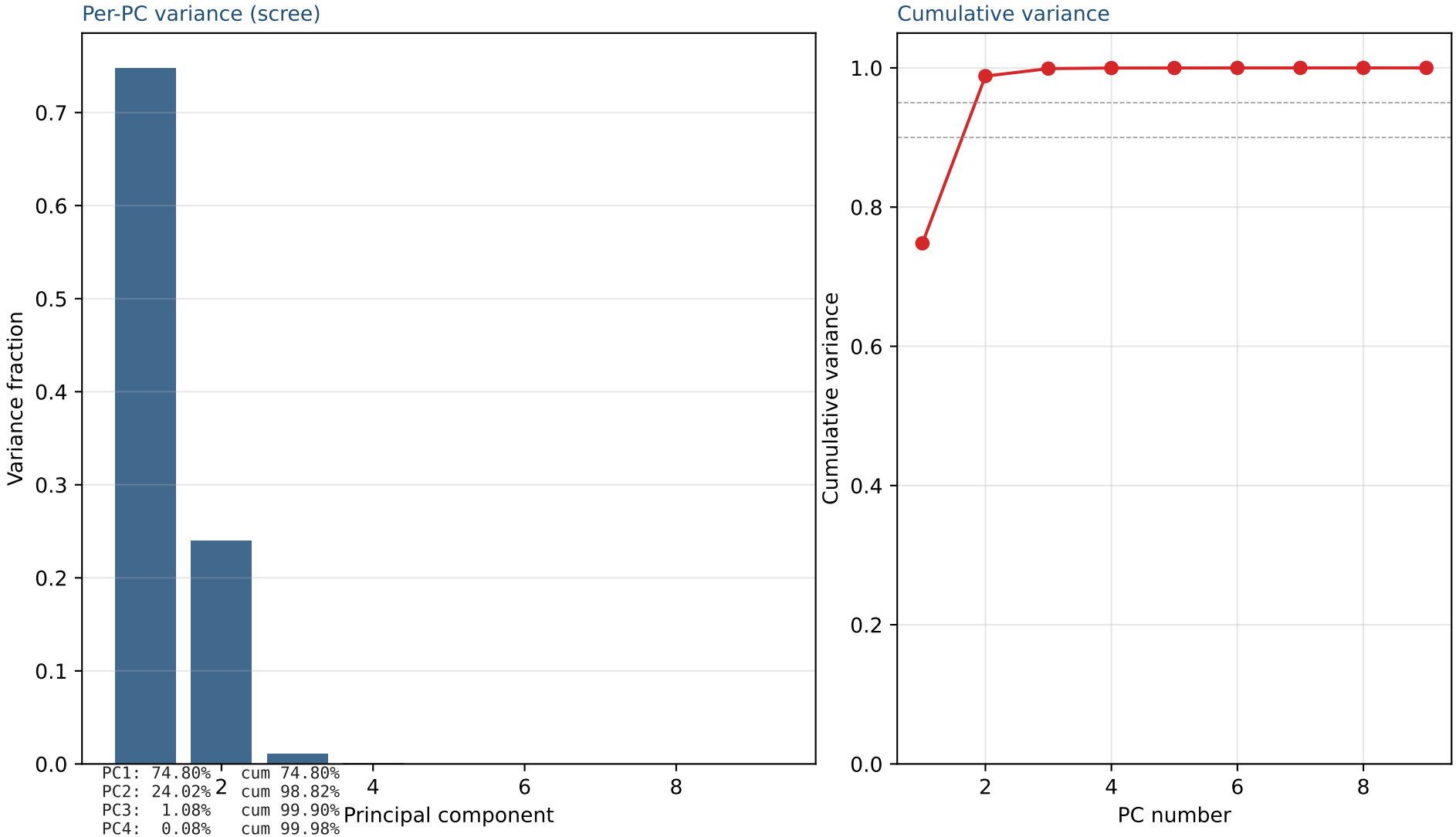


Each ternary closes the chosen 3 carriers; remaining carriers absorbed into closure. ▲ start ■ end | colour map = time index

Triplet source: default heuristic

STAGE 2 | Plate 22 | Scree Plot — CLR PCA

ember_gbr | Engine cnt 2.0.4 | Schema 2.1.0 | Order 2



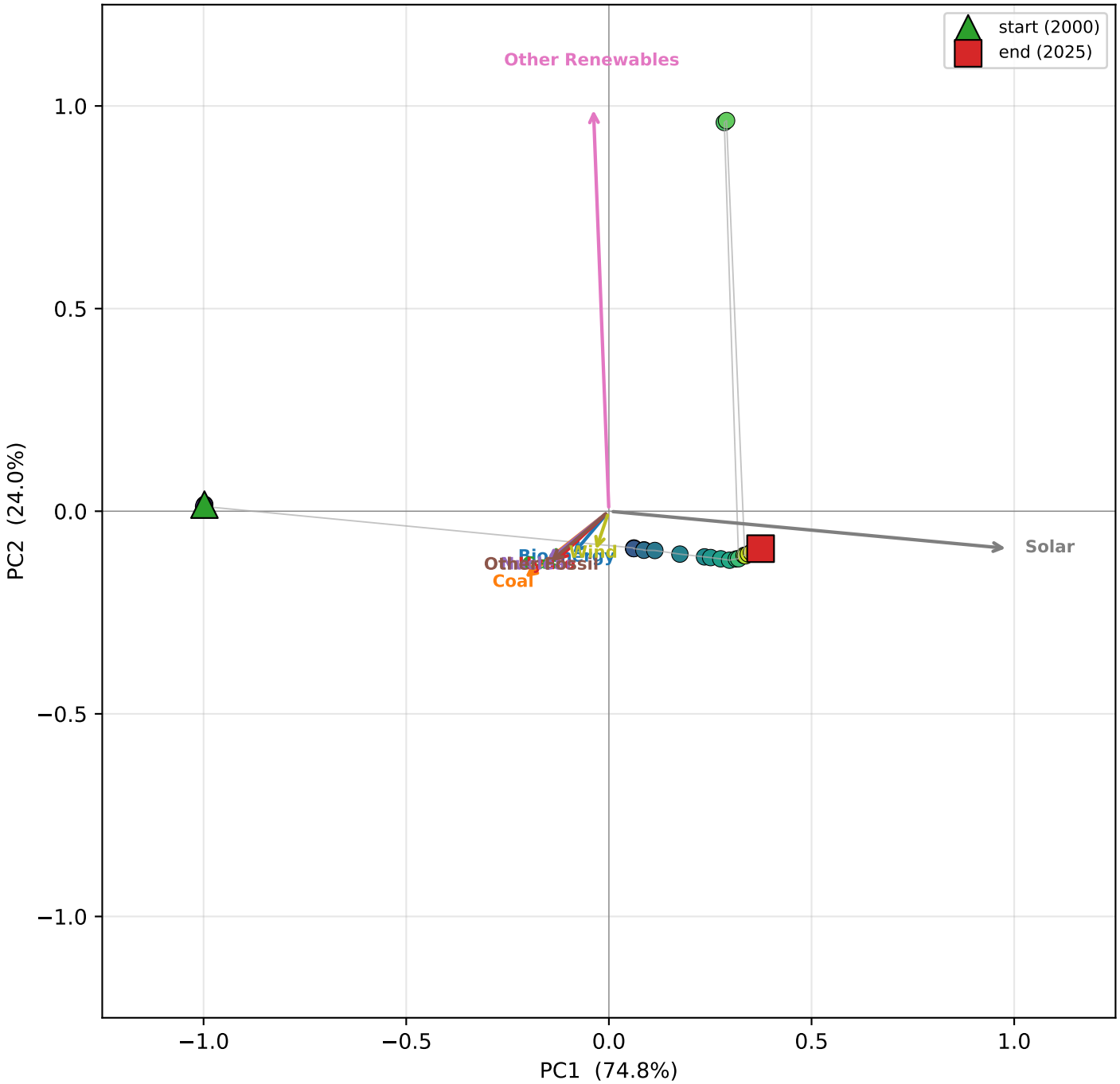

```
PC1: 74.80%   cum 74.80%
PC2: 24.02%   cum 98.82%
PC3:  1.08%   cum 99.90%
PC4:  0.08%   cum 99.98%
PC5:  0.01%   cum 99.99%
PC6:  0.01%   cum 99.99%
... 3 more PCs
```

Scree plot — eigenvalues of the centred CLR covariance matrix. | Computed via SVD of the centred CLR matrix (no reweighting). |

STAGE 2 | Plate 23 | CLR Biplot (CoDa biplot)

ember_gbr | Engine cnt 2.0.4 | Schema 2.1.0 | Order 2

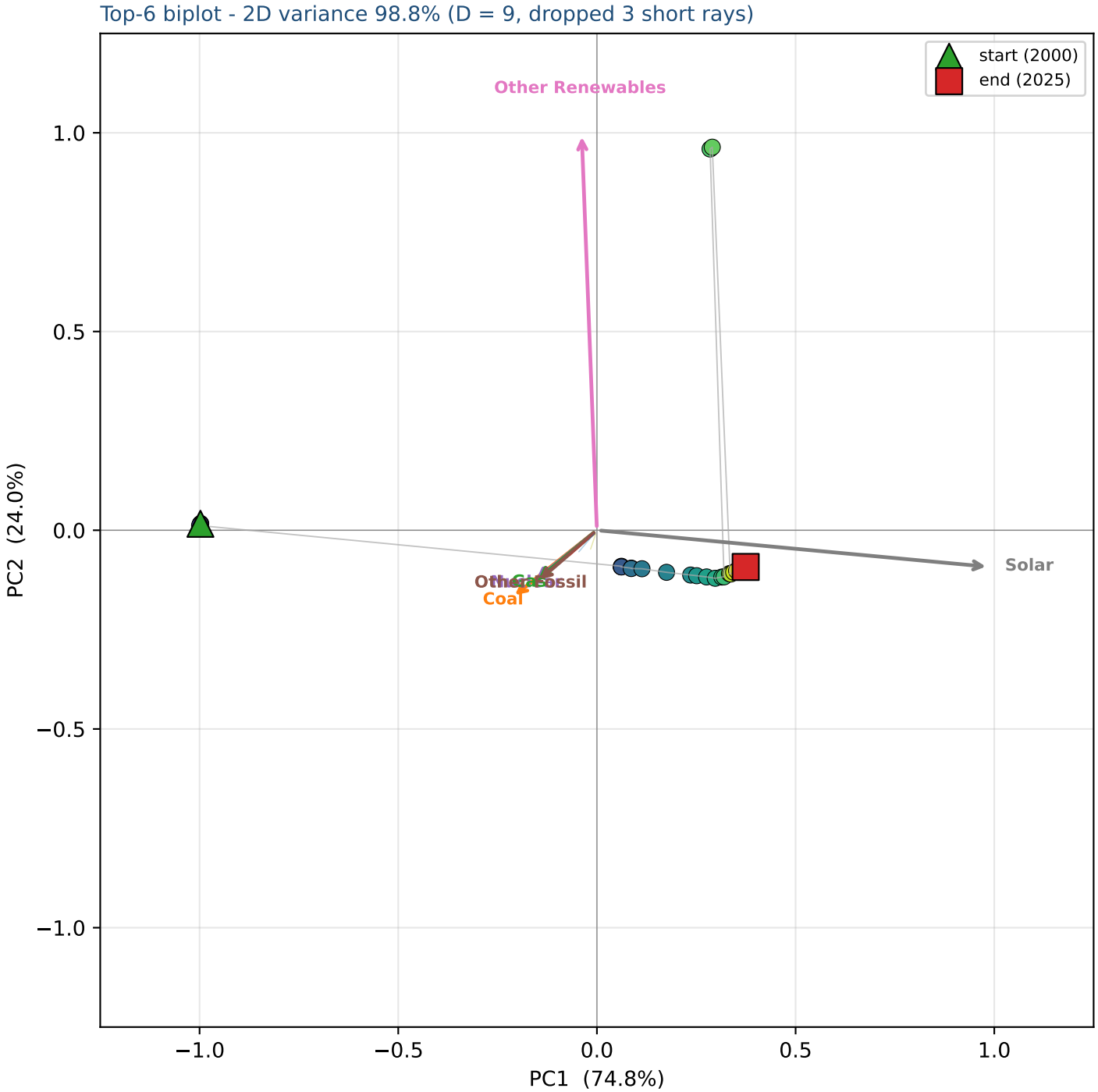
CoDa biplot — covariance form, 2D variance explained 98.8%




```
CoDa biplot (Aitchison-Greenacre):
  - Points: timestep scores in PC(1,2) of c
  - Rays:   carrier loadings (columns of V)
  - Ray length = carrier importance on the
  - Angle between rays ~ negative log-ratio
  - Distance between point projections = ap
    of the Aitchison distance restricted to

PC1 captures 74.8% of variance.
PC2 captures 24.0% of variance.
Total 2D plane:  98.8%
```

STAGE 2 | Plate 24 | CLR Biplot - top-6 carriers by loading magnitude

ember_gbr | Engine cnt 2.0.4 | Schema 2.1.0 | Order 2

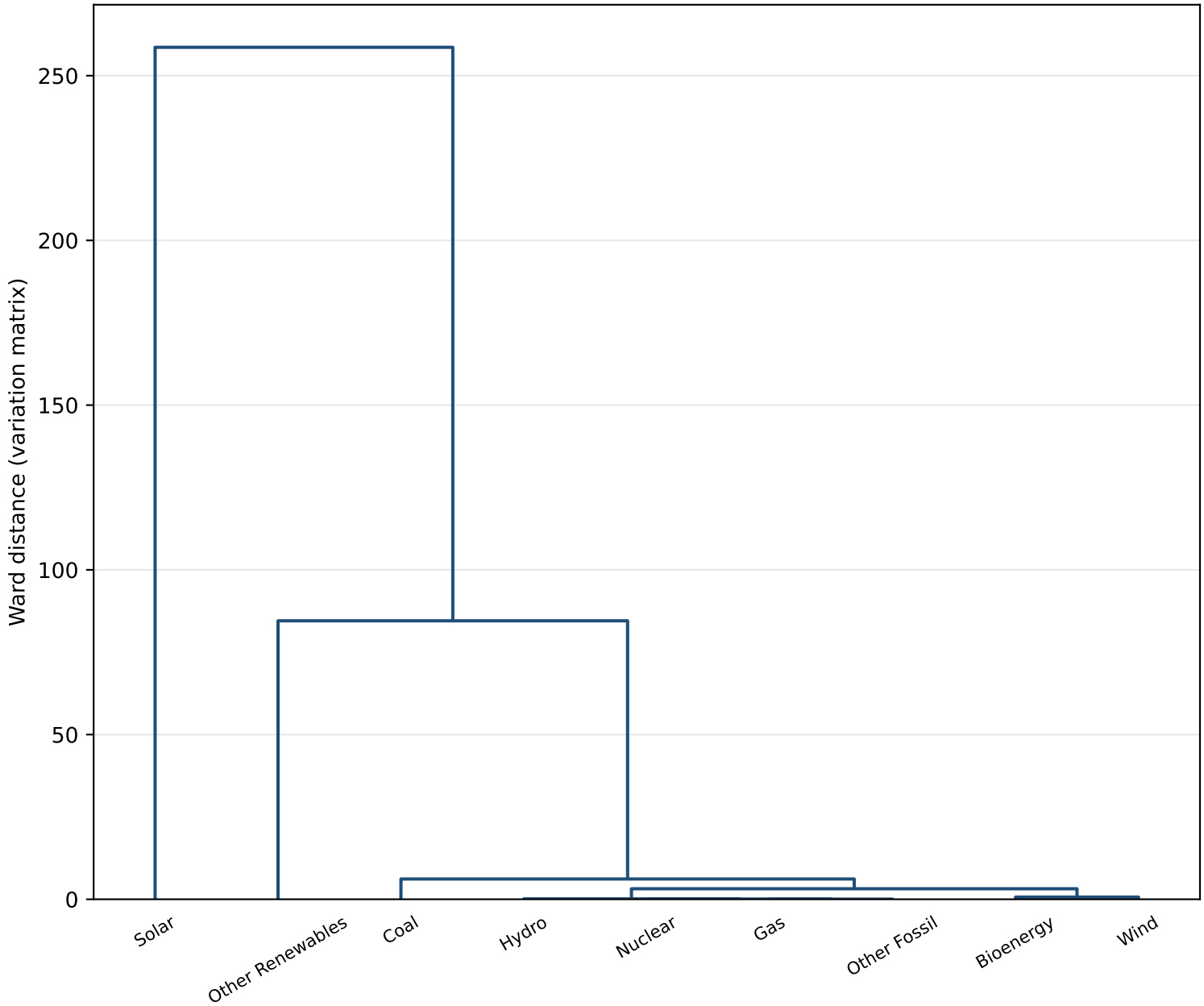



```
Loading magnitudes sqrt(G1^2 + G2^2):

  + Other Renewables 0.9419
  + Solar            0.9353
  + Coal             0.2575
  + Nuclear          0.1949
  + Gas              0.1898
  + Other Fossil     0.1886

Dropped (faint stubs only):
    Hydro            0.1793
    Bioenergy        0.1388
    Wind             0.1036
```

STAGE 2 | Plate 25 | Balance-Dendrogram (Ward on variation matrix)

ember_gbr | Engine cnt 2.0.4 | Schema 2.1.0 | Order 2

Hierarchical balance tree — each merge defines one balance partition




```
Balance-dendrogram (Pawlowsky-Glahn-
  – Ward clustering on the (symmetri
  – Each internal node merges two ca
  – That merge defines a balance:
      b = sqrt(|A||B|/(|A|+|B|)) ·
    where g() is the geometric mean.
  – The balances form an orthonormal

D = 9 carriers → D-1 = 8 balances.
Tightly-clustered carriers (low vari
See companion plate for the full SBP
```

STAGE 2 | Plate 26 | Sequential Binary Partition (SBP) Matrix

ember_gbr | Engine cnt 2.0.4 | Schema 2.1.0 | Order 2

SBP cells: + = numerator, − = denominator, · = not in this balance

	b1	b2	b3	b4	b5	b6	b7	b8
energy	·	·	·	+	−	−	−	−
Coal	·	·	·	·	·	+	−	−
Gas	+	−	−	·	+	−	−	−
Hydro	·	·	+	·	+	−	−	−
uclear	·	+	−	·	+	−	−	−
Fossil	−	−	−	·	+	−	−	−
wables	·	·	·	·	·	·	+	−
Solar	·	·	·	·	·	·	·	+
Wind	·	·	·	−	−	−	−	−

Balance b_k = sqrt(|A_k||B_k|/(|A_k|+|B_k|)) * ln(g(A_k) / g(B_k)) where A_k = numerator set (+ rows), B_k = denominator set (- rows). D = 9; D-1 = 8 bal

```
T = 26,  D = 9
carriers: Bioenergy, Coal, Gas, Hydro, Nuclear, Other Fossil, Other Renewables, Solar, Wind

Navigation (System Course Plot):
  net_distance     = 36.2591 HLR
  path_length      = 100.3591 HLR
  course_directness= 0.3613

ω (deg)        : min 0.00  mean 3.87  max 36.63
Hs             : min 0.2061  mean 0.3159  max 0.4110
Aitchison norm : min 7.9405  mean 35.6606  max 47.0073
κ trace        : min 951591041.27  max 281904362844444423297997601817755648.00
condition #    : min 1.31e+04  max 1.76e+17

Bearings: range [-179.99, 178.70]°,  std 66.18°
Lock events    : 29
Pairs          : 36
|r| > 0.7 pairs: 22

Helmsman frequency (top 5):
  Solar            8  (30.8%)
  Coal             6  (23.1%)
  Wind             5  (19.2%)
  Hydro            3  (11.5%)
  Other Renewables   2  (7.7%)

Variation matrix trace : 0.0000
```